growth is represented by S shaped growth curve.

Biotic potential or Reproductive capacity (r)

It is the maximum reproductive capacity of an organism under optimum environmental conditions.

Carrying capacity (K)

The maximum number of organism that a region can support without environmental degradation is called carrying capacity.

Environmental resistance

Is the sum total of the environmental limiting factors, both biotic and abiotic, which together act to prevent the biotic potential of an organism from being realized.

Table 10.2 Differences between r- selected and K selected species

r selected species (r - Reproductive capacity)	K selected species (K - Carrying capacity)
Smaller sized organisms	Larger sized organisms
Produce many offspring	Produce few offspring
Mature early	Late maturity with extended parental care
Short life expectancy	Long life expectancy
Each individual reproduces only once or few times in their life time	Can reproduce more than once in lifetime
Only few reach adulthood	Most individuals reach maximum life span
Unstable environment, density independent	Stable environment, density dependent

10.11 Population Regulation

The inherent tendency of all animal populations is to increase in number. But it does not increase indefinitely. Once the carrying capacity of the environment is reached, population numbers remain static or fluctuate depending on environmental conditions. This is regulated by many factors which are

1. Density independent – Extrinsic factors
2. Density dependent - Intrinsic factors

Extrinsic factors include availability of space, shelter, weather, food, etc. Intrinsic factors include competition, predation, emigration, immigration and diseases.

10.12 Population Interaction

Organisms belonging to different populations interact for food, shelter, mating or for other necessities. Interaction may be intra specific (interaction within the members of same species) or inter specific (among organisms of different species).

Intra specific association is observed for all livelihood processes like feeding, territoriality, breeding and protection.

Interspecific associations or interactions can be:

Neutral: where different species live together but do not affect each other.

Positive: it is a symbiotic relationship in which no organism in association is harmed and either one or both may be benefitted. It is of two types – Mutualism and Commensalism.

Negative: One or both of the interacting organisms will be affected as in case of competition, predation, parasitism.

The common types of interspecific inter actions are:

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